

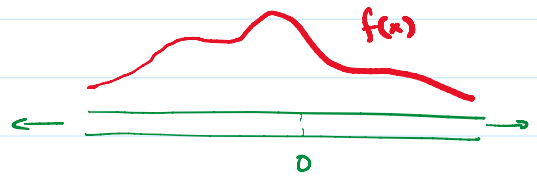
1D Heat Equation for a very long bar

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8- Heat Equation for a very long bar

8.1 The heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$



in general has solution of the form

$$u(x, t; p) = (A \cos(px) + B \sin(px)) e^{-c^2 p^2 t}$$

when we use separation of variables

8.2 In the absence of boundary conditions (such as $u(0, t) = u(L, t) = 0$), we understand that the parameter p can take any value $0 \leq p < \infty$. So then we say:

$$u(x, t) = \int_0^\infty u(x, t; p) dp$$

$$u(x, t) = \int_0^\infty u(x, t; p) dp = \int_0^\infty [A(p) \cos px + B(p) \sin px] e^{-c^2 p^2 t} dp$$

8.3 Remark: Compare this to

$$u(x, t) = \sum_{n=0}^\infty u_n(x, t)$$

$$= \sum_{n=0}^\infty [A_n \cos(\omega_n x) + B_n \sin(\omega_n x)] e^{-c^2 \omega_n^2 t}$$

where ω_n depends on values of $u(0, t)$ and $u(L, t)$ at the boundaries, and $n = 0, 1, 2, \dots$ is discrete.

8.4 Now, if we impose initial condition

$$u(x, 0) = f(x)$$

that is:

$$\int_0^\infty [A(p)\cos(px) + B(p)\sin(px)] dp = f(x)$$

and just like with Fourier coefficients:

$$A(p) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos(px) dx; \quad B(p) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin(px) dx$$

8.5 Remark: Note the usage of the integration limits: $-\infty$ and ∞ . This means we are assuming that the bar spans $-\infty < x < \infty$; a very long bar indeed!

8.6 It can then be shown (see pages 569-570 if interested) that

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(v) \exp\left\{-\frac{(x-v)^2}{4c^2 t}\right\} dv. \quad (**)$$

Furthermore, using new variable $z = \frac{v-x}{2c\sqrt{t}}$, we may

simplify:

$$u(x, t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(x + 2cz\sqrt{t}) e^{-z^2} dz.$$

9- Using Fourier Transform to solve the 1D Heat Equation for a very long bar.

9.1 For any function $f(x)$, we define its Fourier transform as.

∂t

We may now integrate both sides:

$$\int_0^t \frac{1}{\hat{u}} \frac{\partial \hat{u}}{\partial s} ds = \int_0^t -c^2 \omega^2 ds$$

$\omega = \text{const}$ $\omega = \text{const}$

$$\Rightarrow \ln(\hat{u}) = -c^2 \omega^2 t + h(\omega)$$

$$\Rightarrow \hat{u} = P(\omega) e^{-c^2 \omega^2 t}$$

9.4 Since $u(x, 0) = f(x)$
then

$$\mathcal{F}(u(x, 0)) = \mathcal{F}(f(x))$$

$$\Rightarrow \hat{u}(\omega, 0) = \hat{f}(\omega)$$

So from $\hat{u}(\omega, t) = P(\omega) e^{-c^2 \omega^2 t}$

we get: $P(\omega) = \hat{f}(\omega)$

In other words: $\hat{u}(\omega, t) = \hat{f}(\omega) e^{-c^2 \omega^2 t}$

9.5 We may now solve for $u(x, t)$ by simply using the inverse Fourier transform:

$$u(x, t) = \mathcal{F}^{-1}(\hat{u}(\omega, t)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-c^2 \omega^2 t} \cdot e^{i\omega x} d\omega$$

9.6 Now, suppose for $g(x)$, $\mathcal{F}\{g(x)\} = \hat{g}(\omega) = \frac{1}{\sqrt{2\pi}} e^{-c^2 \omega^2 t}$

$$\text{Then } u(x, t) = \int_{-\infty}^{\infty} \hat{f}(\omega) \hat{g}(\omega) e^{i\omega x} d\omega$$

Then, we get the (known) result:

$$u(x, t) = (f * g)(x) = \int_{-\infty}^{\infty} f(p) g(x-p) dp \quad (*)$$

↑
convolution

9.7: We also have: $g(x) = \mathcal{F}^{-1}(\hat{g}(\omega)) = \mathcal{F}^{-1}\left[\frac{1}{\sqrt{2\pi}} e^{-c^2 \omega^2 t}\right]$

$$\Rightarrow g(x) = \frac{1}{\sqrt{2c^2 t} \sqrt{2\pi}} e^{-x^2/(4c^2 t)}.$$

Replacing x with $x - p$ and substituting this into (*) we finally have

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(*)

$$u(x, t) = (f * g)(x) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(p) \exp\left\{-\frac{(x-p)^2}{4c^2 t}\right\} dp.$$

9.8 Example: Let $f(x) = \begin{cases} 100 & \text{if } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$, and let $c=1$

then

$$u(x, t) = \frac{1}{2\sqrt{\pi t}} \int_{-\infty}^{\infty} f(x) e^{-\frac{(x-p)^2}{4t}} dp$$

$$= \frac{1}{2\sqrt{\pi t}} \int_{-1}^1 100 e^{-(x-p)^2/4t} dp$$

$$= \frac{50}{\sqrt{\pi t}} \int_{-1}^1 e^{-(x-p)^2/4t} dp$$

